MILITARY SPECIFICATION

HOSE AND HOSE ASSEMBLY FOR WATER COOLING OF ELECTRONIC EQUIPMENT

This specification is approved for use by all interested Commands of the Department of the Navy and the Marine Corps and is available for use by all other Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a smooth bore, reinforced, seamless hose and end fittings for use in distilled or demineralized water cooling loops in a high voltage environment.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date on invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
QQ-N-281 - Nickel-Copper-Alloy Bar, Plate, Rod, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought.
QQ-S-763 - Steel Bars, Wire, Shapes and Forgings Corrosion-Resisting.

MILITARY
MIL-H-775 - Hose, Hose Assemblies, Rubber, Plastic, Fabric, or Metal (Including Tubing); and Fittings, Nozzles, and Strainers.
MIL-E-9500 - Ethylene Glycol, Technical.

STANDARDS

FEDERAL
FED-STD-162 - Hose, Rubber, Visual Inspection Guide for.
FED-STD-601 - Rubber: Sampling and Testing.

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-177 - Rubber Products, Terms for Visible Defects of.
MIL-STD-278 - Fabrication, Welding and Inspection; and Casting Inspection and Repair for Machinery, Piping and Pressure Vessels in Ships of the United States Navy.
MIL-STD-454 - Standard General, Requirements for Electronic Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D380 - Rubber Hose, Testing.
D876 - Nonrigid Vinyl Chloride Polymer Tubing, Testing.
D2240 - Indentation Hardness of Rubber and Plastics by Means of a Durometer, Test for.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

FSC 4730

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to beginning production a sample shall be examined and tested as specified in 4.3 (see 6.2).

3.2 Material.

3.2.1 Rubber. The tube and cover shall be of ethylene-propylene-diene-monomer (EPDM) or other elastomer. The cover shall be abrasive and ozone resistant and fungus inhibiting.

3.2.2 Hose reinforcement fibers. Hose reinforcement fibers shall be synthetic.

3.2.3 Fittings. End fittings shall be either corrosion-resisting steel in accordance with QQ-S-763, nickel-copper alloy in accordance with QQ-N-281, or nickel-copper-aluminum alloy in accordance with QQ-N-286.

3.3 General construction. The hose shall consist of a flexible, seamless rubber tube, synthetic fiber reinforcement, and seamless rubber cover.

3.3.1 Tube. The inner tube shall have a smooth bore and be of uniform gage. The inner tube shall be free of dirt, mandrel lubricants, and other foreign materials.

3.3.2 Concentricity. Concentricity based on full indicator reading between the inside bore of the hose and the outer surface of the hose shall not exceed the following values: sizes 1/4 inch and under - 0.030 inch; sizes over 1/4 inch up to and including 3/4 inch - 0.040 inch; sizes over 3/4 inch - 0.050 inch.

3.3.3 Dimensions and pressures (hose). Hose dimensions and pressures shall be in accordance with table I as specified (see 6.1.1 and 6.1.2).

Table I - Hose performance characteristics and dimensions.

Hose size	Pressure (pounds per square inch gage (lb/in^2g))			Hose dimensions (inches)				Min. bend radius (inches)	Approximate weight (pound per foot (lb/ft))
	System pressure maximum (max.)	Proof	Burst minimum (min.)	Inside diameter (i.d.) tube		Outside diameter (o.d.) cover			
				min.	max.	min.	max.		
-04	200	400	800	0.250	0.281	0.563	0.625	3.00	0.145
-06	200	400	800	.375	.406	.719	.781	4.00	.190
-08	200	400	800	.500	.531	.844	.906	5.00	.244
-12	200	400	800	.750	.781	1.125	1.187	6.00	.344
-16	200	400	800	1.000	1.047	1.469	1.531	7.00	.634

3.3.4 Temperature range. The hose shall operate through a temperature range of minus 65° to plus 200°F.

3.3.5 Lengths. Unless otherwise specified (see 6.1.1), hoses shall be furnished in the following lengths:

(a) Minimum of 65 percent in lengths 45 to 60 feet.
(b) Maximum of 25 percent in lengths 25 to 44 feet.
(c) Maximum of 10 percent in lengths 3 to 24 feet.
(d) No lengths shall exceed 60 feet.

3.3.6 End fittings. Unless otherwise specified (see 6.1.1 and 6.1.2), the end fittings shall consist of the type and sizes specified on figure 1 and shall be of the reusable, screw-together design. The hose end of the fitting shall be of the solid-socket design capable of meeting the requirements of this specification. The hose end of the reuseable fitting shall be readily removable and reattachable to a new section of hose by the use of commonly available tools normally carried onboard ship. The use of special tools shall not be permitted without cognizant command or agency review. Special tools are defined as those

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tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service (DCAS)).

3.3.6.1 Weights and dimensions. The weight of the end fitting shall be kept to a minimum and the dimensions of the fitting shall fall within the envelope limits shown on figure 1.

3.3.6.2 Finish. End fittings shall be finished smooth and free from defects such as burrs, nicks, and cracks.

3.3.6.3 Welding. Welding, if required, shall be in accordance with MIL-STD-278. Where brazing of fitting parts is used, only silver brazing is permitted. Welding procedures used by the fitting manufacturer, for the materials specified herein, shall be submitted to cognizant command or agency for review. Procedures shall be submitted prior to the start of first article testing.

3.3.6.4 Threads. Where screw-threads are specified on the tube end of the fitting, they shall be in accordance with Handbook H28.

3.3.6.5 Where two parts of the same material are mated together, provisions shall be made to prevent galling and seizing of the mated surfaces. A solid film lubricant, two classes or treatments of the same material, or a combination of any of these methods may be used to prevent galling and seizing.

3.4 Detailed construction.

3.4.1 Hose reinforcement. The reinforcement shall consist of single or multiple plies of synthetic fibers, applied in such a manner as to meet the requirements of this specification. There shall be no broken reinforcing fibers, nor shall any reinforcing fibers be omitted in any length of hose.

3.5 Physical requirements.

3.5.1 Performance. The hose and fittings shall satisfy the physical requirements specified in table I and the following performance requirements:

(a) Examination of product (see 4.5.3).
(b) Assembly (see 4.5.4).
(c) Proof pressure - room temperature (see 4.6.1).
(d) Proof pressure - elevated temperature (see 4.6.2).
(e) Elongation or contraction (see 4.6.3).
(f) Burst pressure - room temperature (see 4.6.4).
(g) Burst pressure - elevated temperature (see 4.6.5).
(h) Cold temperature flexing (see 4.6.6).
(i) Surge pressure (see 4.6.7).
(j) Water circulation - elevated temperature (see 4.6.8).
(k) Electrical resistance (see 4.6.9).
(l) Distilled water immersion (see 4.6.10).
(m) Ethylene glycol - distilled water immersion (see 4.6.11).
(n) Ozone resistance (see 4.6.12).
(o) Distilled water extraction (see 4.6.13).
(p) Concentricity (see 4.6.14).
(q) Hardness (see 4.6.15).
(r) Cold flow resistance (see 4.6.16).
(s) Accelerated aging (see 4.6.17).
(t) Fungus resistance (see 4.6.18).
(u) Bend radius (see 4.6.19).
(v) Water purity (see 4.6.20).

3.6 Marking.

3.6.1 Hose. Unless otherwise specified (see 6.1.1), each length of hose shall have an indelible ink layline along its entire length. This layline shall contain the following information:

(a) Number of this specification.
(b) Hose dash size.
(c) Quarter of year and year of manufacture.
(d) Hose manufacturer's federal supply code (manufacturer's part number and trademark are optional).

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Lettering shall be not less than 3/16-inch high and marking shall be repeated at 9-inch intervals. Marking shall withstand nominal handling, water, shipping and storage; but shall be removable with methyl chloride.

3.6.2 End fittings. End fittings shall have the following information permanently marked into the socket:

(a) Manufacturer's federal supply code or manufacturer's trademark.
(b) Fitting type.
(c) Tube size and hose size.

Example: 55555/A/1/16 where the first five digits refer to the manufacturer's federal supply code and the rest of the markings identify a fitting, 1-inch iron pipe size (ips), for use with a dash 16 hose. The lettering shall be not less than 1/8-inch high. Marking need not appear more than once on each fitting.

3.6.3 Instruction tag. A set of instructions for installing the fittings onto the hose shall be secured to each fitting and shall contain the following:

(a) Steps and tools required to properly attach the fitting to the hose. These steps should be clear and concise.
(b) Any maintenance or assembly precautions considered necessary by the fitting manufacturer.

The instruction tag shall be durable enough so that it does not become illegible or damaged during packaging or storage.

3.7 Technical data. The supplier shall prepare welding procedures, inspection system plan, and first article test reports in accordance with the data ordering documents included in the contract or order (see 6.1.3).

3.8 Workmanship. Hose and hose assembly shall conform to the quality specified herein with no cuts, bruises, pits, sharp edges, etc., shall be clean and smooth, and as specified in 4.5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The supplier shall provide and maintain an inspection system in accordance with the data ordering documents included in the contract or order (see 6.1.3).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination and tests specified in 4.3.3.

4.3.1 Sampling for first article inspection.

4.3.1.1 Hose. For testing of hose only, 17 samples of sizes and lengths shall be provided as specified in table II. Samples requiring fittings shall be provided with end fittings. One-half of the test assemblies shall be made using maximum o.d. hose and one-half shall be made using minimum o.d. hose.

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Table II - Cut hose length (inches) for test hose assemblies.

Hose dash size	Sample assembly numbers (see note 1)							
	1, 2, 3, 4, 5, 6	7 and 8	9 and 10	11 and 12	13	14 and 15	16	17
-08	12.00	20.00	20.00	12.00	20.00 (min.)	3.00	See 4.6.12	2.00
-16	14.00	17.00	17.00	14.00	20.00	3.00	See 4.6.12	2.00

NOTE: 1. Samples 3 through 6 and 9 through 12 are the samples needed when testing end fittings only.

4.3.1.2 End fittings. For testing of end fittings only, 16 end fittings shall be provided. End fitting samples shall be installed on the appropriate samples of hose in the sizes specified in table II.

4.3.1.3 Hose and end fitting combined. In the event that a single manufacturer of both hose and fittings or a hose manufacturer and a fitting manufacturer desire to perform a combined test on the hose and fittings, 17 hose and 24 fitting samples shall be provided with the hose lengths specified in table II.

4.3.2 Test conditions. Test conditions shall be as specified in 4.3.2.1 through 4.3.2.3 unless otherwise specified in 4.3.3.

4.3.2.1 Test fluids. Test fluids shall be triple-distilled water or a mixture of ethylene glycol and triple-distilled water.

4.3.2.2 Test pressures. Pressures shall have a tolerance of plus or minus 10 percent.

4.3.2.3 Test temperature. Tests shall be conducted with the ambient and fluid temperature at 70° to 90°F.

4.3.3 First article inspection. Test samples shall be identified by number as indicated in table III and subjected to the examination and tests in the sequence specified in table III. If only end fittings are being tested, the elongation of contraction test (see 4.6.3), the cold temperature flexibility test (see 4.6.6), and the electrical resistance test (see 4.6.9) need not be performed.

Table III - Sequence of first article inspection.

Examination or test	Sample number and examination and test sequence																
	1/ 1 min.	2 max.	3 min.	4 min.	5 max.	6 max.	7 min.	8 max.	9 min.	10 max.	11 min.	12 max.	13	14	15	16	17
Examination of product (see 4.5.3)	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Assembly (see 4.5.4)	2	2	2	2	2	2	2	2	2	2	2	2	-	-	-	-	-
Proof pressure - room temperature (see 4.6.1)	3	3	3	3	3	3	3, 5	3, 5	3	3	3	3	-	-	-	-	-
Proof pressure - elevated temperature (see 4.6.2)	-	-	4	4	4	4	-	-	-	-	-	-	-	-	-	-	-
Elongation or contraction 2/ (see 4.6.3)	4	4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

See footnotes on next page

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Table III - Sequence of first article inspection (con).

Examination and test	Sample number and examination and test sequence																
	1/ 1 min.	2 max.	3 min.	4 min.	5 max.	6 max.	7 min.	8 max.	9 min.	10 max.	11 min.	12 max.	13	14	15	16	17
Burst pressure - room temperature (see 4.6.4)	5	5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Burst pressure - elevated temperature (see 4.5.5)	-	-	5	5	5	5	-	-	-	-	5	5	-	-	-	-	-
Cold temperature flexing 2/ (see 4.6.6)	-	-	-	-	-	-	4	4	-	-	-	-	-	-	-	-	-
Surge pressure (see 4.6.7)	-	-	-	-	-	-	-	-	4	4	-	-	-	-	-	-	-
Fluid circulation - elevated temperature (see 4.6.8)	-	-	-	-	-	-	-	-	-	-	4	4	-	-	-	-	-
Electrical resistance 2/ (see 4.6.9)	-	-	-	-	-	-	-	-	-	-	-	-	4	-	-	-	-
Distilled water immersion 2/ (see 4.6.10)	-	-	-	-	-	-	-	-	-	-	-	-	-	3	-	-	-
Ethylene glycol - distilled water immersion 2/ (see 4.6.11)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2	-	-
Ozone resistance 2/ (see 4.6.12)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2	-
Distilled water extraction 2/ (see 4.6.13)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2
Concentricity 2/ (see 4.6.14)	-	-	-	-	-	-	-	-	-	-	-	-	2	-	-	-	-
Hardness 2/ (see 4.6.15)	-	-	-	-	-	-	-	-	-	-	-	-	3	-	-	-	-
Cold flow resistance 2/ (see 4.6.16)	-	-	-	-	-	-	-	-	-	-	-	-	5	-	-	-	-

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Table III - Sequence of first article inspection (con).

Examination and test	Sample number and examination and test sequence																
	1/ 1 min.	2 max.	3 min.	4 min.	5 max.	6 max.	7 min.	8 max.	9 min.	10 max.	11 min.	12 max.	13	14	15	16	17
Accelerated aging 2/ (see 4.6.17)	-	-	-	-	-	-	-	-	-	-	-	-	6	-	-	-	-
Fungus resistance 2/ (see 4.6.18)	-	-	-	-	-	-	6	-	-	-	-	-	-	-	-	-	-
Bend radius 2/ (see 4.6.19)	-	-	-	-	-	-	-	-	-	-	-	-	-	2	-	-	-
Water purity (see 4.6.20)	-	-	-	-	-	-	-	6	-	-	-	-	-	-	-	-	-

1/ Minimum and maximum refer to o.d. hose (see 4.3.1.1).

2/ Not required when testing fittings only.

4.3.3.1 First article test report. The supplier shall prepare a first article test report in accordance with the data ordering documents included in the contract or order (see 6.1.3).

4.4 Quality conformance inspection.

4.4.1 Sampling.

4.4.1.1 Lot.

4.4.1.1.1 Hose. A lot shall consist of all units of hose of the same size offered at one time for purposes of quality conformance inspection. One unit of hose is equal to each 2-foot increment of length of hose offered for quality conformance inspection.

4.4.1.1.2 Fittings. A lot shall consist of all fittings of the same size offered at one time for the purpose of quality conformance inspection.

4.4.1.2 Sampling for visual and dimensional examination.

4.4.1.2.1 Hose. A random sample of hose units shall be selected from each lot (see 4.4.1.1.1) in accordance with MIL-STD-105 for normal level of examination using an acceptable quality level (AQL) value of 1.5 percent. Examination shall be conducted as specified in 4.5.1.

4.4.1.2.2 Fittings. A random sample of fittings shall be selected from each lot (see 4.4.1.1.2), in accordance with the master table of MIL-STD-105 for normal level of examination using an AQL value of 1.5 percent. Fittings shall be examined as specified in 4.5.2.

4.4.1.2.3 Hose assemblies. Hose assemblies shall be visually and dimensionally examined to insure conformance with the requirements of this specification.

4.4.1.3 Sampling for tests.

4.4.1.3.1 Hose. A random sample of hose, in 18-inch lengths shall be selected from each lot in accordance with table IV. A 2-inch piece shall be cut from one end of each 18-inch sample and shall be used in conducting the electrical resistance test specified in 4.6.9. The 16-inch sample shall be fitted with fittings which have been previously tested and accepted, and shall be subjected to the elongation and contraction test specified in 4.6.3 and to the burst test specified in 4.6.4. In addition, all hose shall be proof tested in accordance with 4.6.1 except that the installation of fittings onto bulk hose is not required.

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Table IV - Sampling of hose for tests.

Number of units in lot	Number of 18-inch lengths in sample
1 to 3	0
4 to 15	1
16 to 75	2
76 to 300	3
301 to 1300	4
1301 to 3200	6
3201 to 8000	8
8001 and greater	12

4.4.1.3.2 Fittings. A random sample of fittings shall be selected from each lot in accordance with table V and shall be subjected to the proof and burst tests specified in 4.6.1 and 4.6.4. In performing these tests, the fittings shall be fitted onto hose lengths (not less than 15 inches in length) which have been previously tested. Fittings used for first article testing shall not be offered for delivery.

Table V - Sampling of fittings for tests.

Number of fittings in lot	Number of fittings in sample
1 to 2	0
3 to 110	2
111 to 300	4
301 to 500	6
501 to 1300	10
1301 and greater	14

4.4.1.3.3 Hose assemblies. Prior to make-up of hose assemblies, samples of hoses and fittings shall be selected and tested in accordance with 4.6.3 and 4.6.4 to insure conformance with the requirements of this specification. After completing these tests, hose assemblies shall be made-up and proof tested in accordance with 4.6.1.

4.5 Visual and dimensional examinations.

4.5.1 Hose. Each of the hose samples selected in accordance with 4.4.1.2.1 shall be visually and dimensionally examined to determine conformance with table I. Any sample hose containing one or more major visual or dimensional defects shall be rejected and if the number of defective hose samples exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. The evaluation of defects shall be in accordance with FED-STD-162. Terms for visible defects shall be as defined in MIL-STD-177.

4.5.2 End fittings. Each sample fitting selected in accordance with 4.4.1.2.2 shall be visually and dimensionally examined to determine conformance with 3.3.7. Any sample fitting containing one or more dimensional defects shall be rejected, and if the number of defective fittings exceed the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.5.3 Examination of product. Hose and fittings shall be examined to determine conformance to this specification with respect to sizes and workmanship.

4.5.4 Assembly. The hose shall accept the application of applicable end fittings (see 3.3.7) without undue effort or cutting of inner tubes when assembled as recommended.

4.5.4.1 There shall be no evidence of collapsing or excessive bulging of the inner tube at the nipple ends when examined visually after assembly.

4.6 Tests.

4.6.1 Proof pressure test - room temperature. Test sample shall be proof pressure tested to the value specified in table I for a period of not less than 30 seconds and not more than 2 minutes. Any evidence of leakage from hose or fittings shall constitute failure.

4.6.2 Proof pressure test - elevated temperature. Test fluid shall be circulated at a temperature of 200° ± 5°F through the test sample for 15 minutes. The circulation shall then be stopped and the assembly pressurized to the proof pressure value specified in table I within 30 seconds, for a period of not less than 60 seconds and not more than 2 minutes. Any evidence of leakage from hose or fittings shall constitute failure.

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4.6.3 Elongation or contraction test at room temperature. Test sample shall be held in a straight unpressurized condition and a 10-inch standard length marked off on the hose. It shall then be pressurized to the operating pressure specified in table I for 30 seconds minimum. At the end of this 30-second period and while still pressurized, the standard length shall be measured and recorded. The percentage of change in the original measured length shall not exceed plus 2 percent to minus 4 percent for all dash sizes.

4.6.4 Burst pressure test - room temperature. One end of the test sample shall be attached to the pressure source. The test sample shall be extended straight and the free end shall not be restrained or fastened in any way. Air must be excluded from the hose bore and free end sealed. The burst pressure specified in table I shall be attained within 25 + 5 seconds. The hose shall not burst, fittings shall not blow off or loosen, and there shall be no leakage from the hose or fittings at pressures below the burst pressures specified in table I.

4.6.5 Burst pressure test - elevated temperature. Test fluid shall be circulated at a temperature of 200° + 5°F through the test sample for 15 minutes. Circulation shall then be stopped and the assembly pressurized to the applicable burst pressure specified in table I within 25 + 5 seconds. The hose shall not burst, fittings shall not blow off or loosen, and there shall be no leakage from the hose or fittings at pressures below the burst pressures specified in table I.

4.6.6 Cold temperature flexing. This test requires mandrels with radii equal to the minimum bend radii specified in table I for each size hose being tested. Test sample shall be filled with a coolant consisting of a mixture of 35 percent distilled water and 65 percent refined grade ethylene glycol (by volume), and held holding at minus 65° to minus 70°F for 24 hours. While holding, the assemblies shall be laid out in a straight position. After aging and while still at the specified temperature, the test sample shall be tested by flexing it about its respective mandrel circumference through 180°, and then bending it in the reverse direction about the mandrel circumference. Each bend shall be completed at a substantially uniform rate in 12 + 3 seconds. After flexing, the test sample shall be allowed to warm to room temperature and visually examined for cover cracks. The test sample shall then be proof pressure tested as specified in 4.6.1. Evidence of cover cracks or leakage shall be cause for rejection.

4.6.7 Surge pressure test. Test fluid shall be circulated at a temperature of 200° + 5°F through the test sample for 15 minutes. The circulation shall then be stopped and the assembly quickly pressurized to 300 lb/in^{2}g for 5-second duration and then released. This surge pressure cycle shall be repeated 100 times. Note: After each ten surge cycles, circulate the fluid through the sample for 3 minutes to bring it up to temperature and then continue with the surge cycles. Evidence of rupture prior to completion of the 100 cycles shall constitute failure.

4.6.8 Fluid circulation - elevated temperature. Test fluid, maintained at a temperature of 200 + 5°F, shall be circulated through the test sample for a period of 168 + 1 hours. Fluid pressure shall be maintained at 200 lb/in^{2}g throughout the test period. Evidence of leakage from hose or fittings shall constitute failure.

4.6.9 Electrical resistance test. The electrical resistance of the hose shall be determined by testing a 1-inch sample of hose as follows:

(a) Place the hose sample between two contact plates with the ends of the sample held firmly against the plates. Make sure that hose ends are cut square and smooth to assure a complete contact of hose ends and plates. Provisions shall be made for preventing corona during testing.
(b) Apply a voltage differential of 1000 direct current (d.c.) across the contact plates and measure the current.
(c) The hose shall be nonconducting with a minimum electrical resistance of 500 megohms per inch for all sizes.

4.6.10 Distilled water immersion. Test sample shall be immersed for 21 days in distilled water. Procedure and preparation of test specimens shall be in accordance with ASTM D380. The volume increase of the tube shall not exceed 10 percent; weight change shall not exceed 10 percent; and the durometer change shall not exceed plus or minus 5 percent.

4.6.11 Ethylene glycol - distilled water immersion. Test requirements shall be as specified in 4.6.10 except the sample shall be immersed in a coolant consisting of a mixture of 35 percent distilled water and 65 percent (by volume) technical grade ethylene glycol in accordance with MIL-E-9500; no contaminants shall be added. The volume increase of the tube shall not exceed 10 percent, weight change shall not exceed 10 percent, and the durometer reading change shall not exceed plus or minus 5 points.

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4.6.12 Ozone resistance. Three specimens shall be die-cut from the outer cover of untested hose and prepared according to method 4111 of FED-STD-601. Each specimen shall be placed in a fixture in such a manner that the specimen shall be held at room temperature for 45 minutes, then placed in an ozone chamber for 168 hours at a temperature of 100° ± 2°F and exposed to an ozone concentration maintained at 50 ± 5 parts of ozone per one millionth parts of air. The test specimens shall be examined every 24 hours for cracking with 7 power magnification. The first observable cracking shall be recorded. After 168 hours of testing, specimens shall exhibit no cracking when examined under 7 power magnification.

4.6.13 Distilled water extraction test. A 10 gram (g) sample of the inner tube consisting of a single rectangular shaped portion of the compound buffed to approximately 1/16-inch thickness shall be immersed completely in a reflux condensor containing not less than 35 milliliter (ml) and not more than 70 ml of distilled water. After 1 hour of boiling and allowing to cool to room temperature, the distilled water shall be poured off and the bottle and sample washed clean with 25 ml more of distilled water. These portions shall be combined, evaporated to dryness in a weighed glass flask on a water bath or electric hot plate, dried in an air oven for 1 hour at 220°F, and weighed. A blank test shall be conducted in which the same amount of distilled water shall be evaporated and dried in the same manner, and a correction applied to the weight determined above. Nonvolatile matter extracted shall not exceed 2 percent, by weight.

4.6.14 Concentricity. The concentricity of the hose shall be measured in accordance with method 2031 of FED-STD-601 and shall meet the requirements of 3.3.3.

4.6.15 Hardness. The hardness shall be measured by use of the Shore A Durometer test of ASTM D2240.

4.6.16 Cold flow resistance. The hose shall first be proof pressure tested as specified in 4.6.1. After testing, the specimen shall be doubled back on itself and the ends tied together with heat-resistant tape. The specimen shall be subjected to a temperature of 200°F for 24 hours in circulating air. At the end of 24 hours of heat exposure, the specimen shall be allowed to cool for 30 minutes and shall again be proof pressure tested. A flexible rod with rounded (beveled) edge whose diameter is 80 percent that of the hose i.d. shall be inserted into the hose. This rod must pass easily through the hose without lubrication.

4.6.17 Accelerated aging. Resistance of the hose to accelerated aging shall be determined as specified in method 7001 of FED-STD-601. After aging, test samples shall be proof pressure (see 4.6.1) and burst pressure (see 4.6.4) tested.

4.6.18 Fungus resistance. Resistance to fungus shall be determined as specified in ASTM D876. No fungus growth shall be observed.

4.6.19 Bend radius. Each specimen shall be bent to the minimum bend radius specified in table I. A flexible rod whose diameter is nine-tenths of the hose i.d. shall be inserted. The rod must pass easily through the hose.

4.7 Inspection of preparation for delivery. The packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation-packaging, packing, and marking. Hose, hose assemblies, and reusable end fittings shall be preserved-packaged level A or C, packed level A, B, or C as specified (see 6.1.1 and 6.1.2), and marked in accordance with MIL-P-775.

5.2 Marking. In addition to any special marking required by the contract or order (see 6.1.1 and 6.1.2), or herein, interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data.

6.1.1 Bulk hose and end fittings. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of hose (see 3.3.4).
(c) Lengths of hose (see 3.3.6).
(d) End fittings size (pipe end and hose) and quantity (see 3.3.7).
(e) Hose marking, if other than as specified (see 3.6.1).
(f) Levels of preservation-packaging and packing required (see 5.1).
(g) Special marking required (see 5.2).

6.1.2 Hose assemblies. When the ordering activity specifies hose assemblies, the procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of hose (see 3.3.4).
(c) End fitting size (pipe end and hose) (see 3.3.7).
(d) Type of configuration.
(e) Length of free length of hose legs.
(f) Preservation-packaging and packing required (see 5.1).
(g) Special marking required (see 5.2).

6.1.3 Contract data requirements. When this specification is used in a procurement invoking the data requirement clause of the Armed Services Procurement Regulations (ASPR) paragraph 7-104.9(n) and which incorporates a DD Form 1423 Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified in the cited Data Item Description (DID) and delivered in accordance with such CDRL. When the ASPR provisions are not invoked, the data specified below shall be delivered in accordance with the contract requirements.

Specification paragraph	Data requirements	Service	Applicable DID	Options
3.3.7.3	Procedure welding	SH	UDI-H-23383	-----
4.1.1	Inspection system plan	SH	UDI-R-23574	-----
4.3.3.1	Report, first article inspection	SH	UDI-T-23450	-----

(Copies of DID's required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

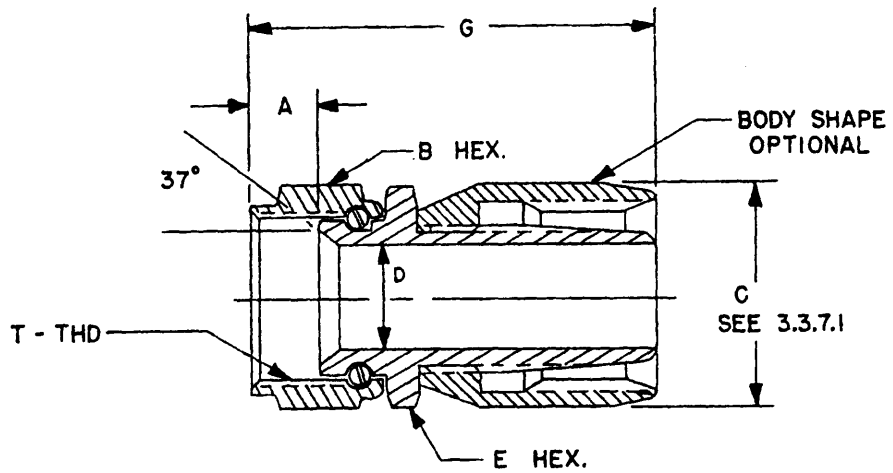
6.2 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

User activities:
Navy - EC, CG

Preparing activity:
Navy - SH
(Project 4730-N486)

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SH 10346

O.d. tube size	Hose size	T Thread CL-2B	A (NOM)	B HEX (NOM)	C (MAX)	D (MIN)	G (MIN)	E HEX (NOM)
1/4	-04	0.4375-20 UNCF	0.368	0.56	0.76	0.175	2.025	0.562
1/4	-06	.4375-20 UNJF	.368	.56	.97	.170	2.151	.56
3/8	-06	.5625-18 UNJF	.380	.68	.97	.294	2.233	.68
1/2	-06	.7500-16 UNJF	.434	.88	.97	.294	2.397	.88
3/8	-08	.5625-18 UNJF	.380	.68	1.11	.294	2.264	.68
1/2	-08	.7500-16 UNJF	.434	.88	1.11	.388	2.397	.88
5/8	-08	.8750-14 UNJF	.504	1.00	1.11	.388	2.575	1.00
3/4	-08	1.0625-12 UNJ	.568	1.25	1.11	.388	2.622	1.25
3/4	-12	1.0625-12 UNJ	.568	1.25	1.45	.606	2.922	1.25
1	-12	1.3125-12 UNJ	.600	1.50	1.45	.606	3.096	1.50
1	-16	1.3125-12 UNJ	.600	1.50	1.79	.841	3.534	1.50
1-1/4	-16	1.6250-12 UNJ	.640	1.81	1.79	.841	3.629	1.81

NOTES: 1. Swivel may be attached to nipple by a metal upset process (crimping) as an alternate.
2. All dimensions in inches.

Figure 1 - Joint industrial conglass (JIC) swivel.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

EXTRUDED FINISHES

TUBES ALUMINUM-ALLOY

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